



Dwight Look College of

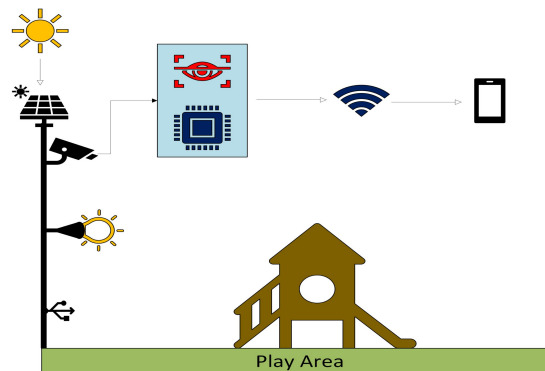
ENGINEERING
TEXAS A&M UNIVERSITY

Team 65: Solar Playground with Security System Bi-Weekly Update 1

**Team Members: Zayan Prasla, Alex Allahar, Jackson
Kitchel, Victor Igbokwe
Sponsor: Wonhyeok Jang
TA: Zien Wang**

Project Summary

A solar-powered security system with a motion-detecting camera. This system provides near real-time video feeds and send automated alerts for suspicious activity through a mobile app, enhancing safety in these areas.



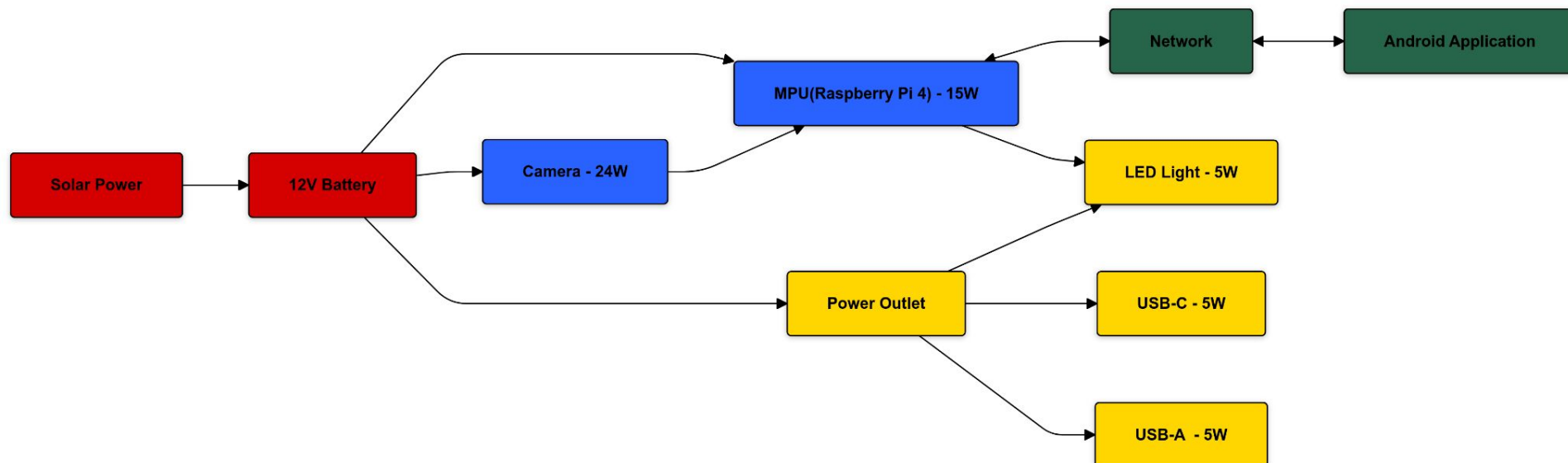
Subsystem Overview

Red - Jackson Kitchel (Power Generation)

Yellow - Victor Igbokwe (Power Distribution)

Blue - Alex Allahar (Microprocessor & Sensors)

Green - Zayan Prasla (Network & Application)



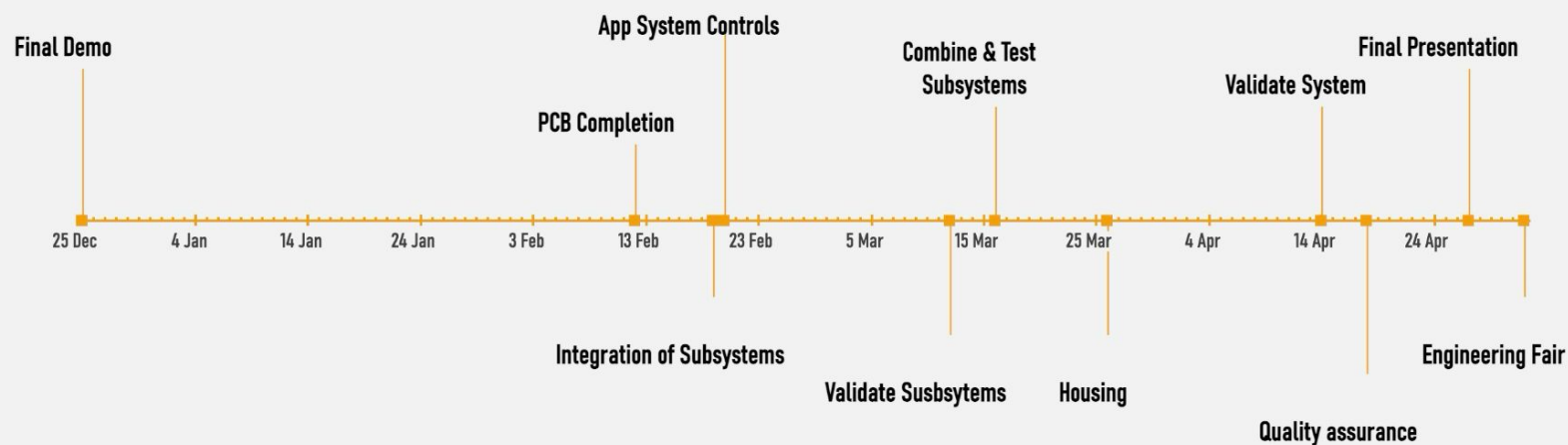


Major Project Changes for 404

- Removed Object Detection
- LED Deterrence
- Adding Cooling



Project Timeline





Power Generation

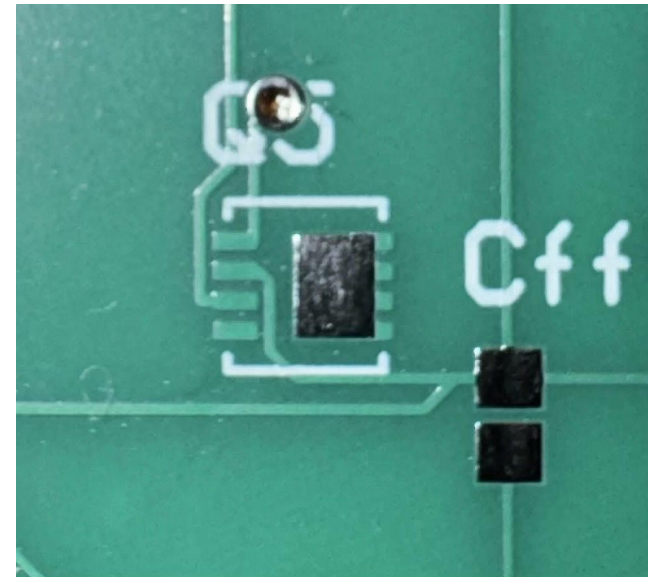
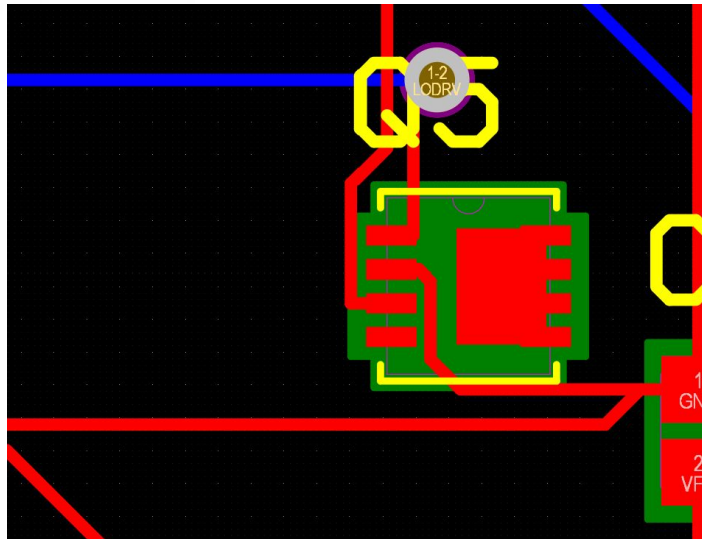
Jackson Kitchel

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
PCP has been updated to address issues that were found in 403. Parts and PCB has been ordered.	- Solder and test new PCB upon arrival. - Integrate with other subsystems.

Power Generation

Jackson Kitchel

- Problem with footprint and missing trace to be fixed with new order





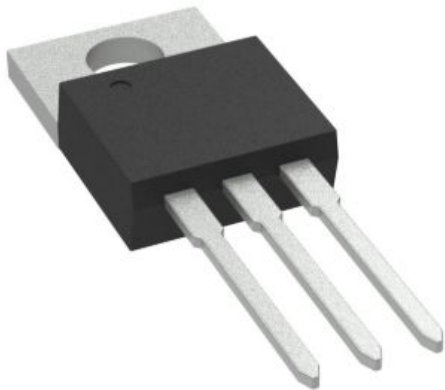
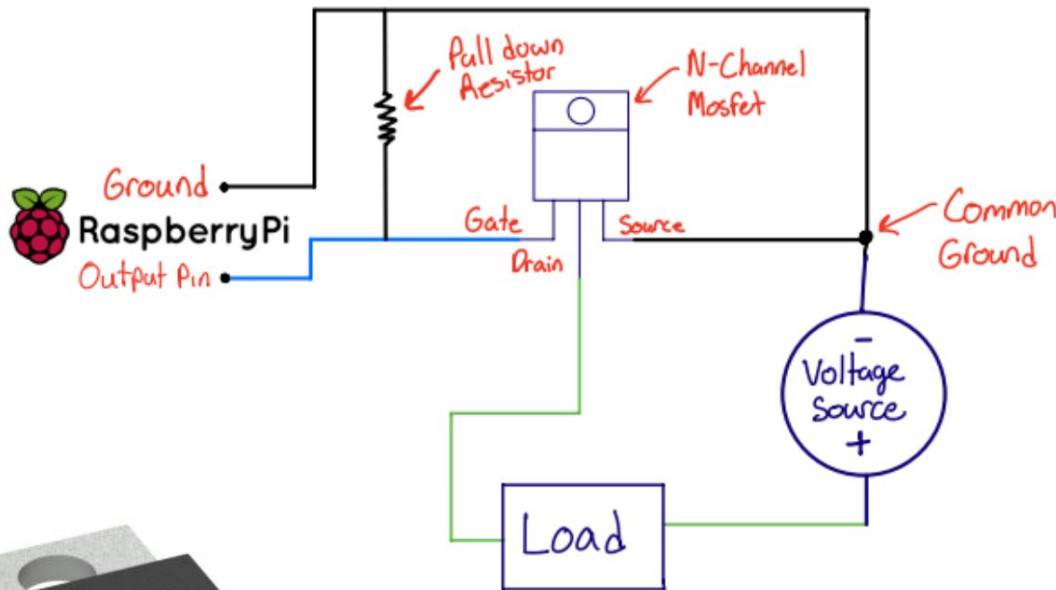
Power Distribution

Victor Igbokwe

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
Buck Converter system experienced issues due to complexity of IC chip that. New PCB has been designed and ordered using a Texas Instruments IC chip.	- Soldering and testing my new buck converter system. - Design and order a new PCB that will implement using an N-Channel Mosfet as a switch to control the Security light.

Power Distribution

Victor Igbokwe



Still Developing:

Using N-channel as a switch

- Common Ground between microcontroller and Power supply
- Pull-down resistor at the output pin.
- Setting the output pin to High will allow current flow while setting it to low will disrupt current flow.



Microprocessor & Sensors

Alex Allahar

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
- Tested and Validation CSI Camera	- Integration of Firebases - Designing the cooling and housing for MPU, Camera, & Sensors.



Microprocessor & Sensors

Low-Power Mode

Works

- GPIO Sensors
- Automated Video Capture
 - Fixed / Variable
- Thumbnails Generation
- Local Event History
- Cloud Video Storage

Streaming Mode

Works

- Real-Time Live Stream
- Live Stream Web Server
- OpenCV Motion Detection
- App Notification

Still Developing

- User Activation via App



Subsystem 4 Application

Zayan Prasla

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
- Removed fixed event history examples from 403 and connected it to Firebase for live updates. - Added reset password for the login page and connected it via backend to the user's specific email.	- App to microprocessor connection via Firebase - Camera and LED controls - Live stream - Constantly improving frontend UI as well



Subsystem 4 Application

Zayan Prasla

Backend/Security

Works

- User Login and Account Creation
 - Password Policy Enforced
 - User accounts stays logged in until manually logged out
- Email Verification
 - Verification link expires after 5 minutes
- Internet Connection
- Password Reset
- Email Address Change
- Event History connected to Firebase Realtime Database

Still Developing

- App to Microprocessor connection
- New Sign-in Methods
- SMS Verification
- 2-step verification

Frontend

Works

- Login, Create Account, Home, Settings, and Event History pages all work
- SOS button
- Error messages for when password policy is not properly followed or
- User friendly UI

Still Developing

- Improvements to the home page UI is needed
- Camera and LED buttons are visible but not yet functional
- Charging Statistics

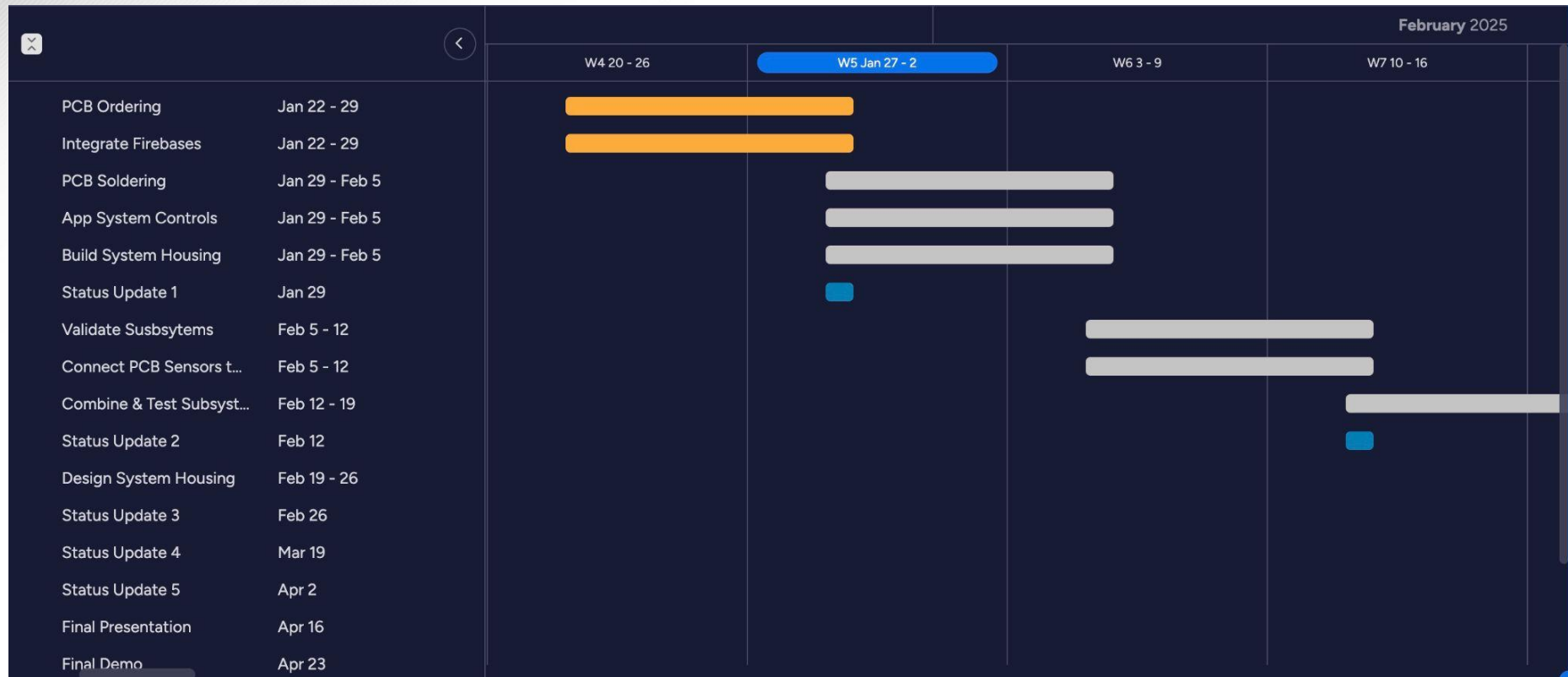


Parts Ordering Status

[illegible]



Execution & Plan



Thank You